

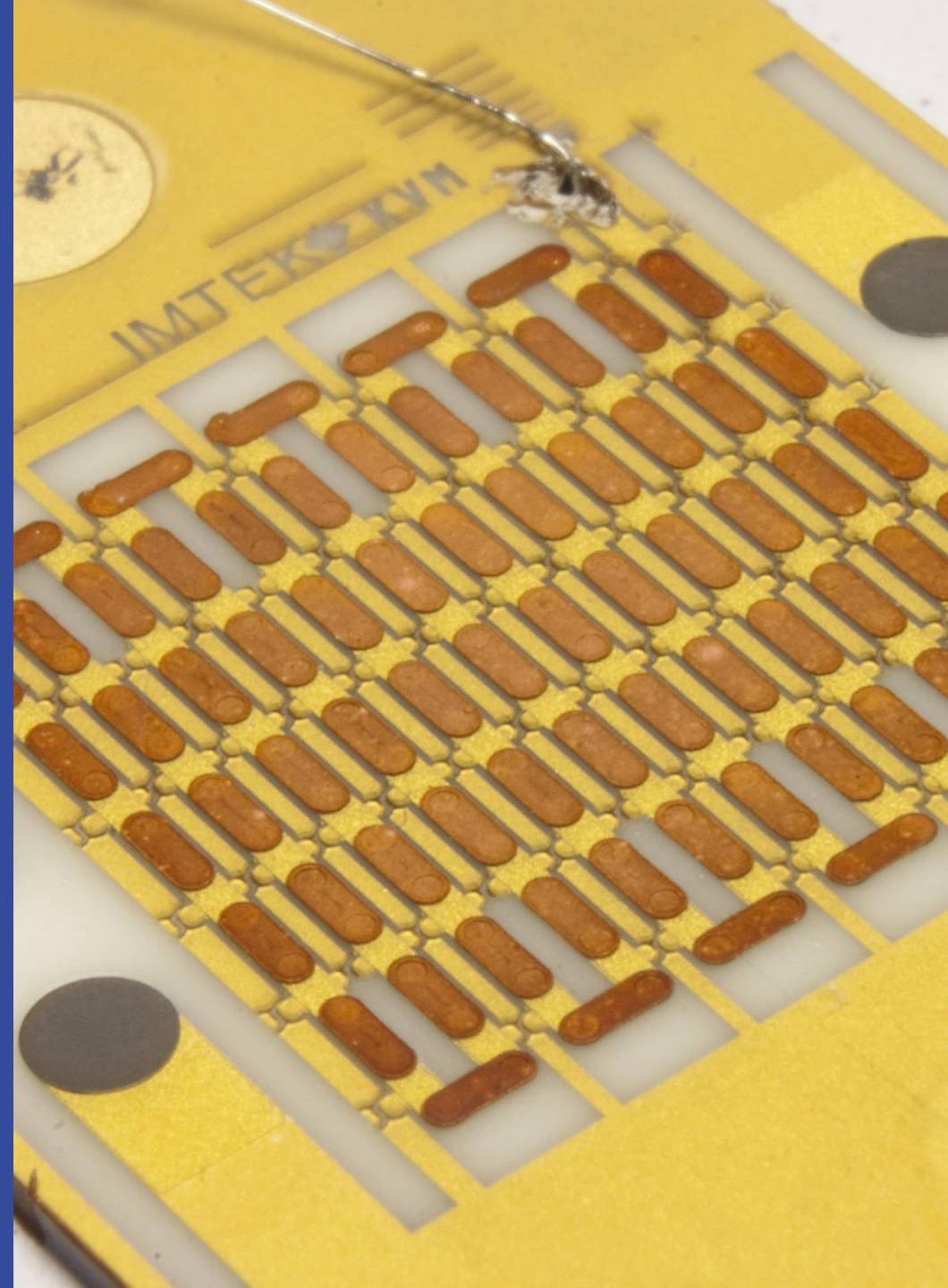
Micro Energy Harvesting

Chapter 9: Power Management

Micro Energy Harvesting SoSe 2026

IMTEK – Lehrstuhl für Konstruktion von Mikrosystemen

Prof. Dr. Peter Woias, Dr. Suman Kundu, Dr. Frank Goldschmidtboing

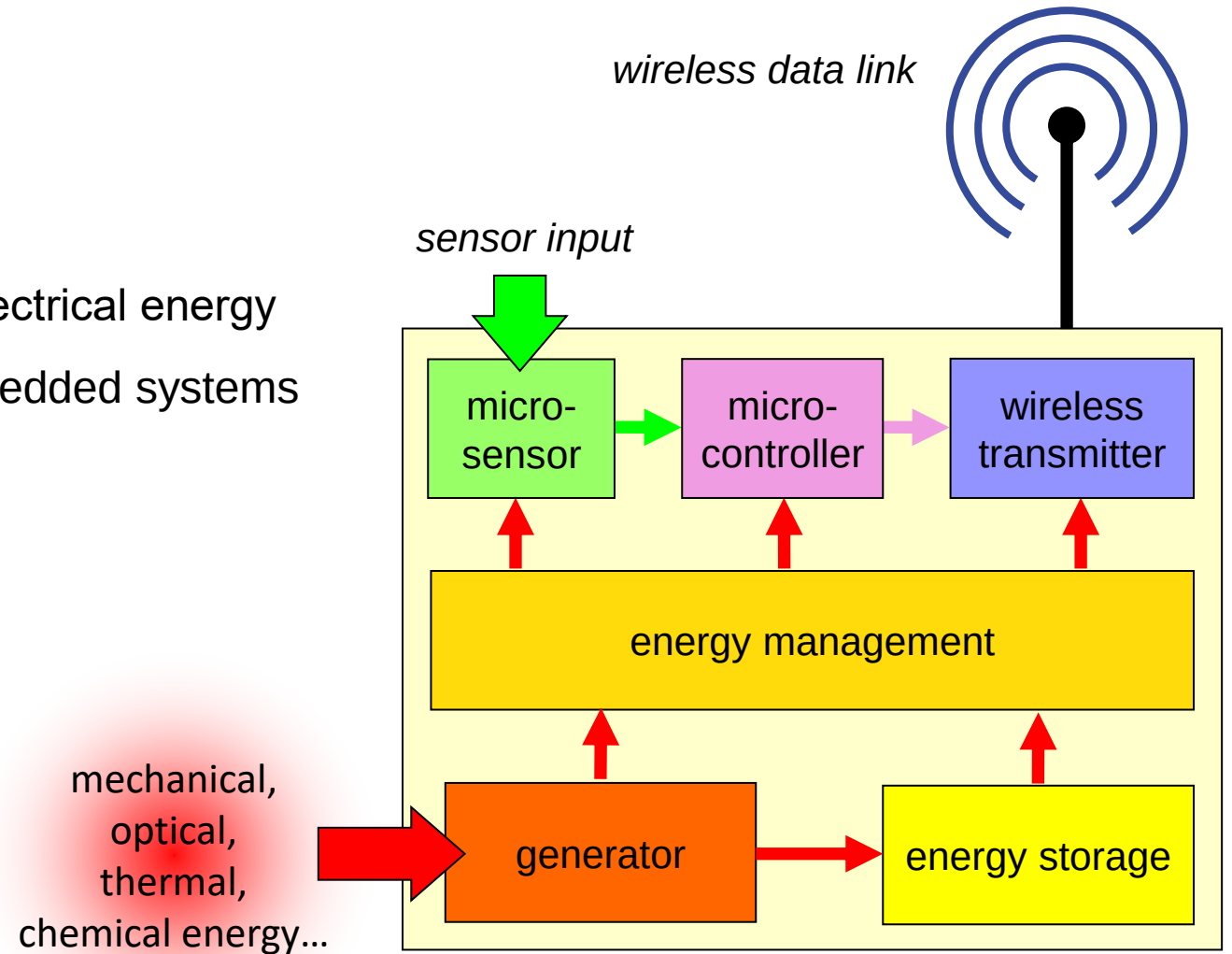


Electronic Power Management

Motivation

Our most famous graph in MEH...

- conversion of local „ambient“ energy into electrical energy
- Goal: energy-autonomous operation of embedded systems
- One generator
- One power source
- One storage element



Electronic Power Management

Different Energy Characteristics

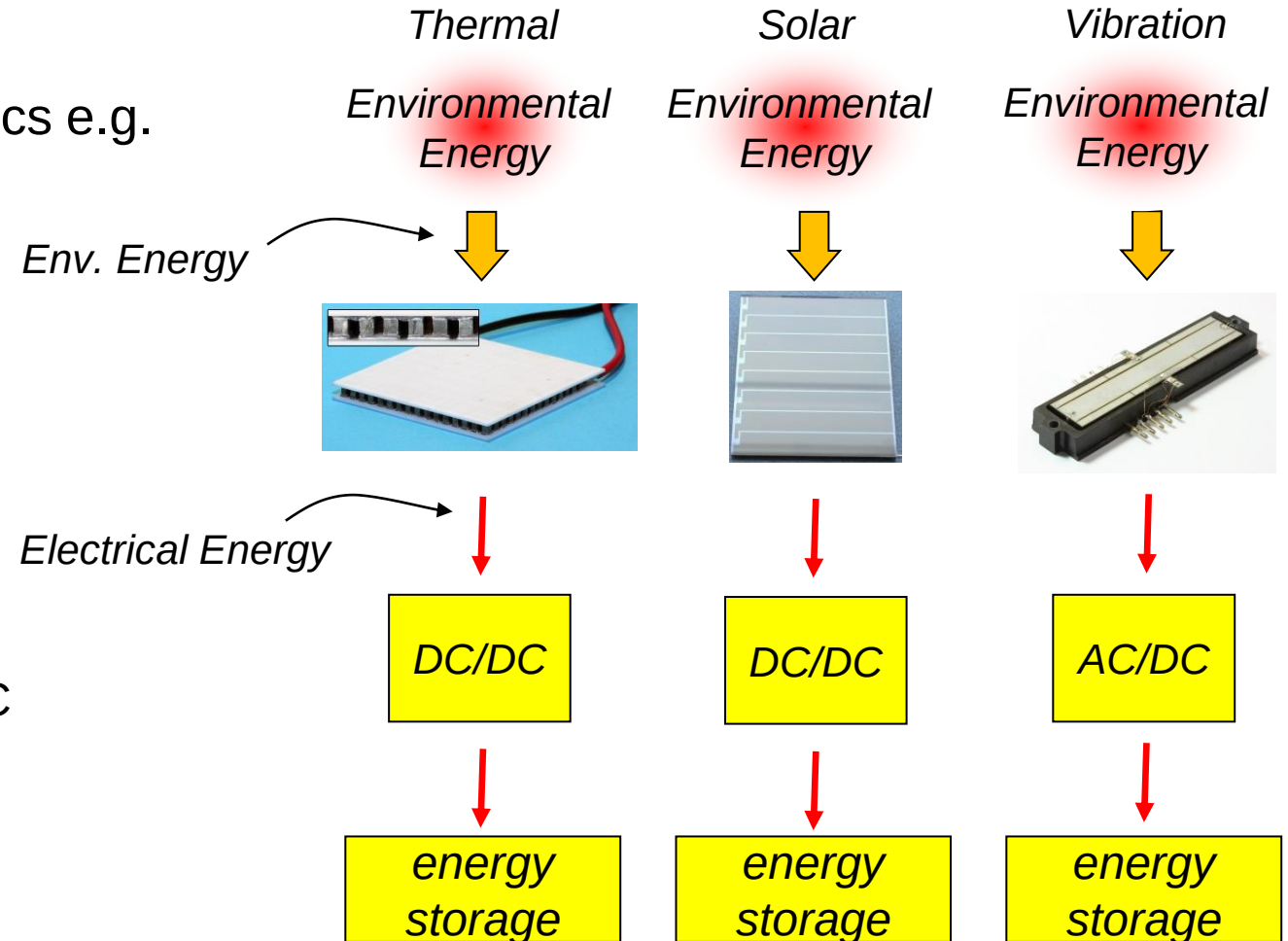
Different sources have different characteristics e.g.

- Thermoelectrics
- Solar cells
- Piezoelectric vibration harvesting

Would it be possible to use the same type of storage?

- Thermo: Very low voltage, low current, DC
- Solar: Reasonable voltage, high current, DC
- Piezo: High voltage, low current, AC

➔ **Power conversion is necessary**
(rectifier and transformer)



Electronic Power Management

Different Storage Concepts

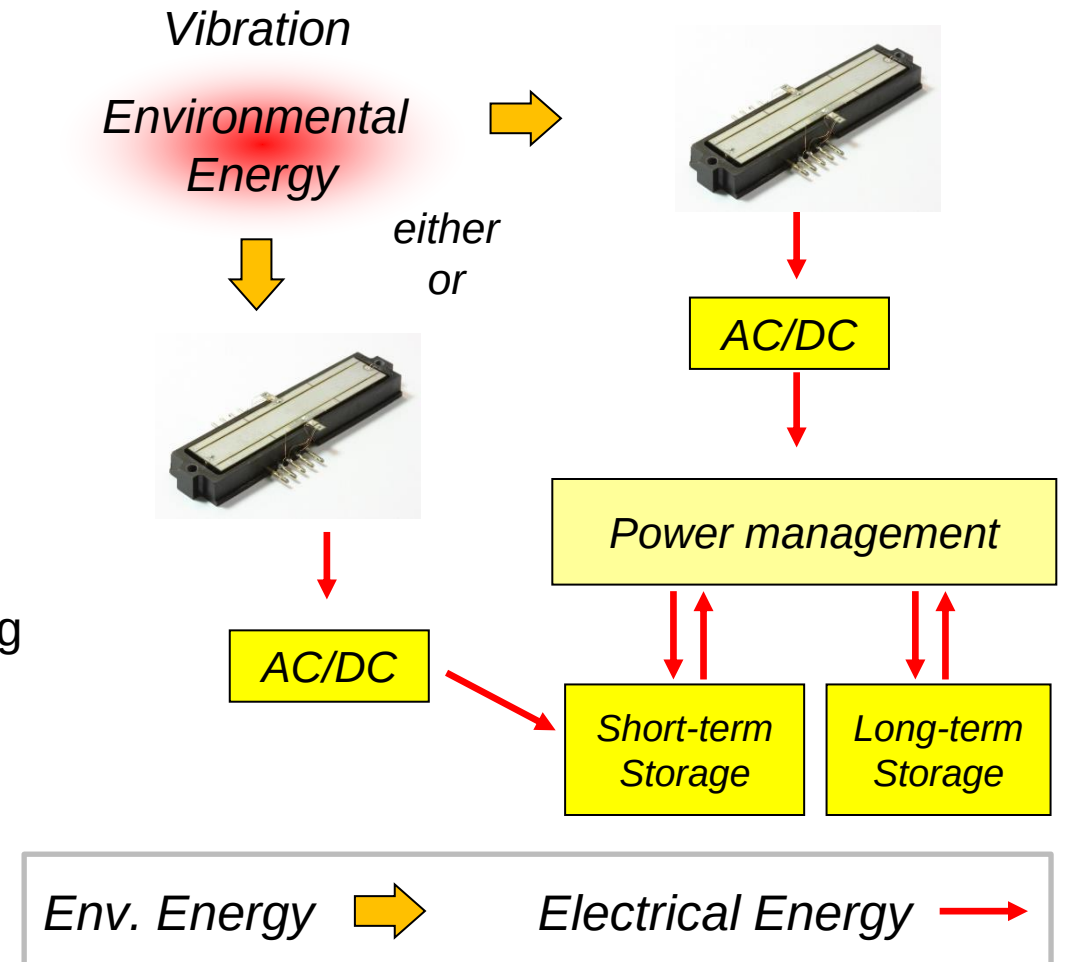
There is typically more than one „storage“ used

- Short-term storage (high leakage, high power), i.e. a buffer capacitor
- Long-term storage (low leakage, low power), i.e. a super cap or a small battery

What are the characteristics of my storage ?

- Capacity
- (Min/ Max) voltages & current for charging & discharging
- Charge retention rate (leakage currents)
- Power output capabilities

➔ A management for power shaping is needed to fit the different storages (U/I)



Electronic Power Management

Different Voltages

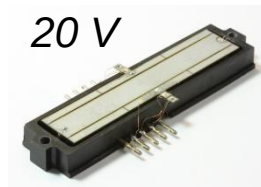
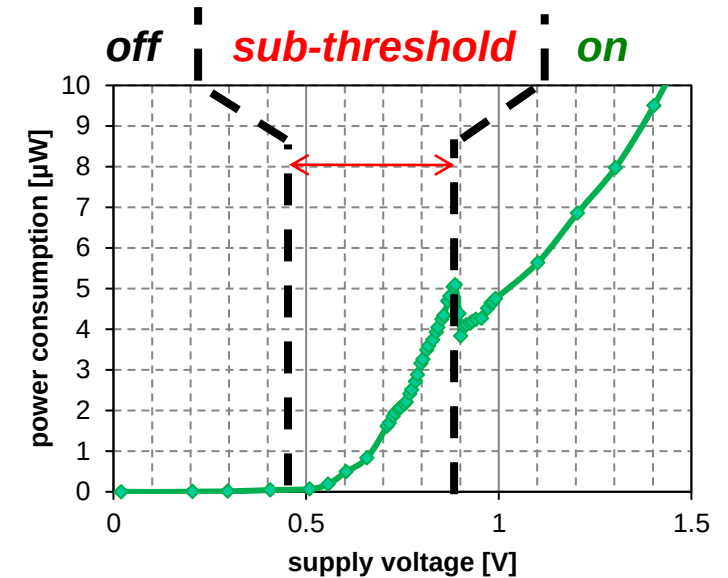
What to do with over/ undervoltages?

- Black-out / Cold-start (from 0V)
 - Hysteresis switch
 - Start-up buffer
- Brown-out (undervoltage, sub-threshold voltage, e.g. 0.6 V)
 - Hysteresis switch
 - Start-up buffer
- Overvoltage / Overpower (Piezo, Pyro,...)
 - Buck-conversion
 - Fuses (get rid of the abundant energy in a resistor)

➔ Power management to operate the electronics within their specifications



Voltcraft (50K)
© Voltcraft



20 V

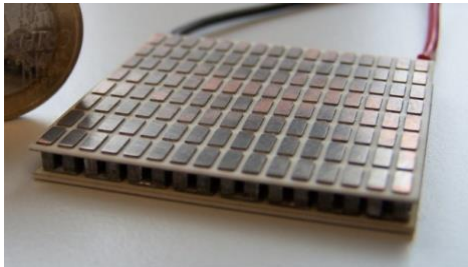


6,3 V max.



Low-voltage Step-up Converters Requirements

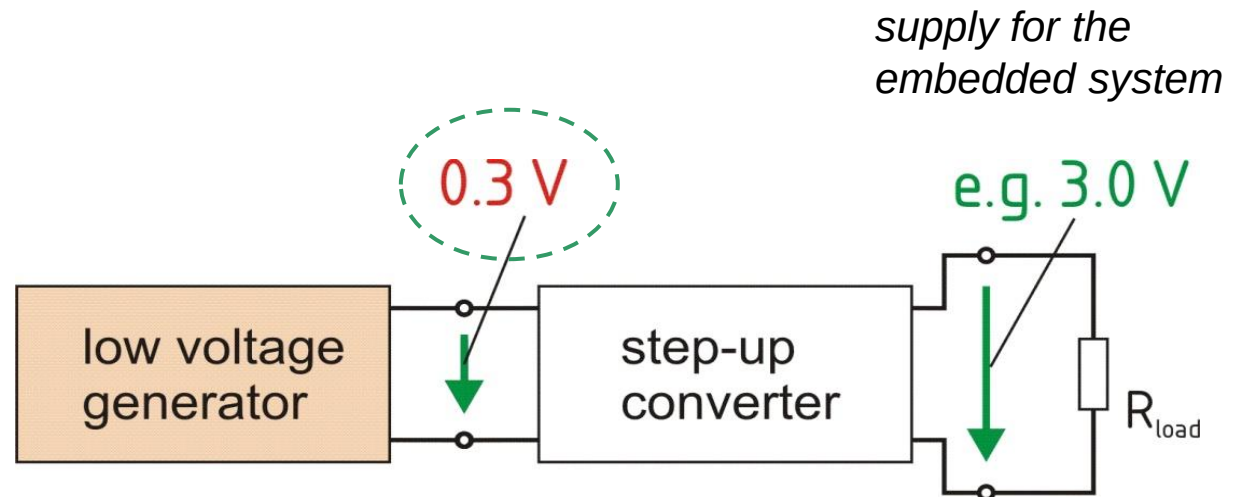
- upward conversion of low input voltages (several mV) into high output voltages (several V)
- high voltage conversion ratio (up to 100...200)
- low start-up voltage of the converter (10 mV 1 V)
- sufficient conversion efficiency (30% and better)
- minimal power losses



TEG
© Peltron



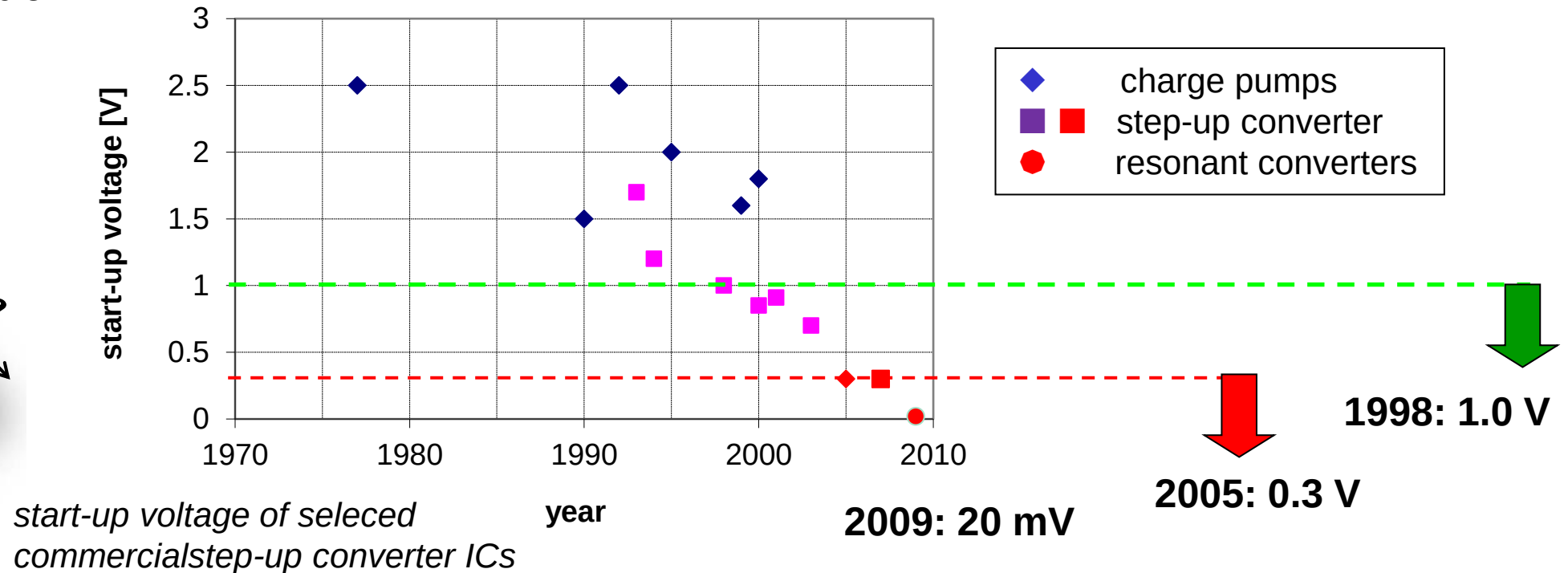
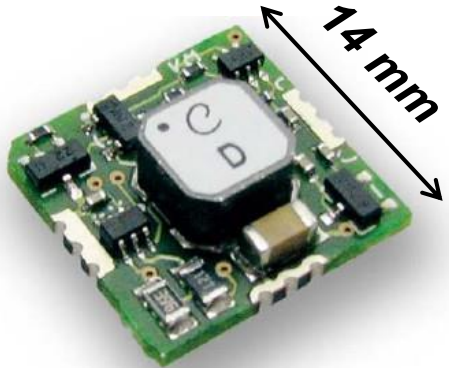
solar cell
(under low light)



(Commercial) Low-voltage Step-up Converters

Start-up Voltage

- level of the input voltage, when the converter starts operation
- may be defined for no-load ($R_{\text{load}} \rightarrow \infty$) at the output or, more realistic, for loading conditions
- only few ICs available with $V_{\text{start}} < 0.3 \text{ V}$



Low-voltage Step-up Converters

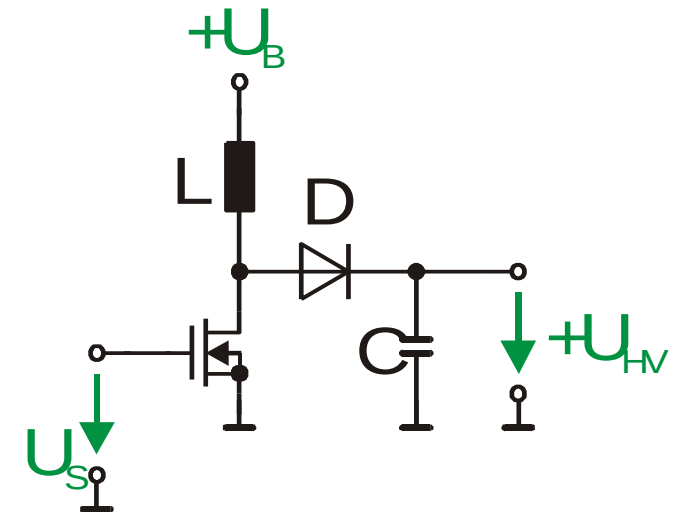
Inductive Step-up Converters (aka „boost converter“)

Advantages for Energy Harvesting

- simple circuit
- allows the generation of positive or negative output voltages
- good to acceptable efficiency (at input voltages above 1 V)
- available as monolithic IC

Problems and Design Challenges

- only a few commercial devices with low start-up voltages ($< 0.3 \text{ V}$)
- bad efficiency at „light load“



*boost converter for
positive output voltages*

Low-voltage Step-up Converters

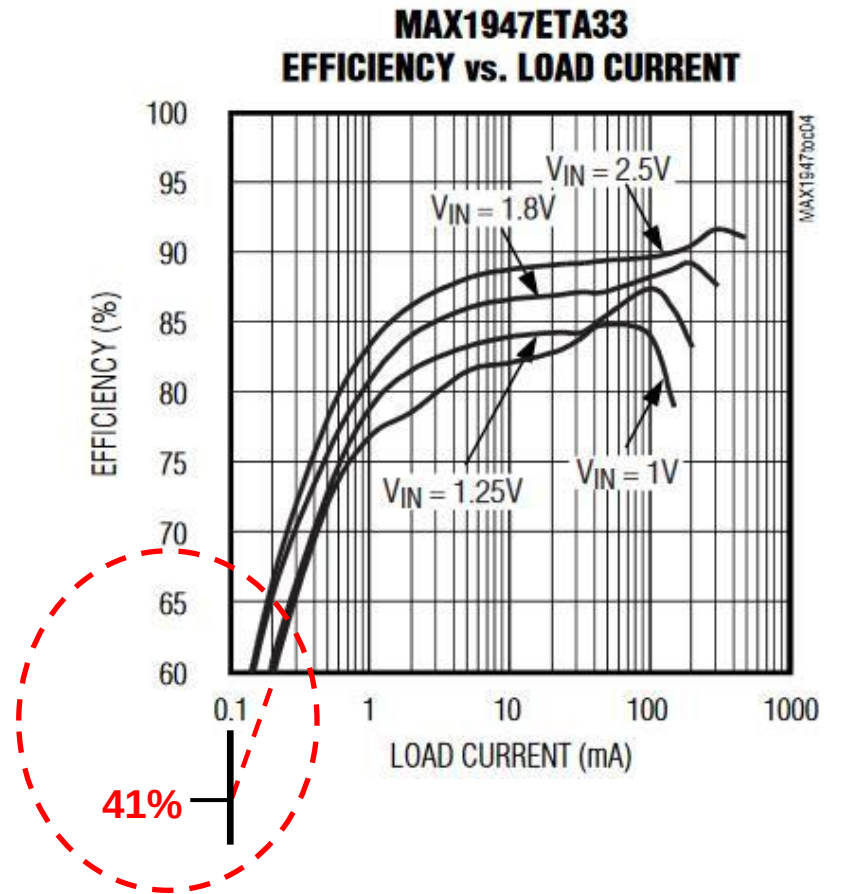
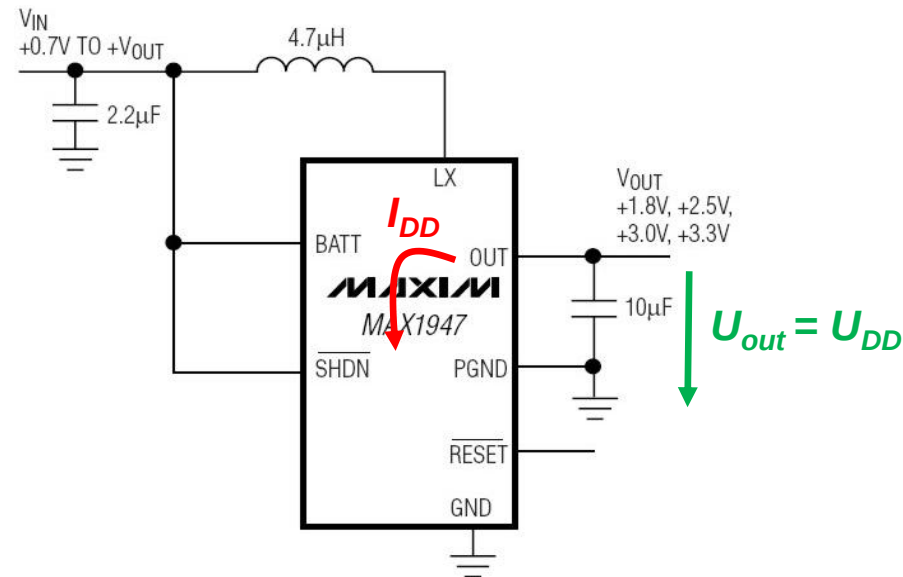
Start-up Voltage and Power Losses

Example: Maxim MAX 1947ETA3.3

- start-up voltage: $U_{in,min} = 0.7 \text{ V}$ (at zero load)
... $= 0.8 \text{ V}$ ($R_L = 1 \text{ k}\Omega$)

- shutdown current: $I_{SD} = 2 \mu\text{A}$
- min. supply current: $I_{DD,min} = 70 \mu\text{A}$
- important:

once started, the chip is internally supplied from the higher output voltage U_{out}



**estimated bad „light load“
efficiency at $I_L = 0.1 \text{ mA}$
(output power: $330 \mu\text{W}@3.3\text{V}$)**

Low-voltage Step-up Converters

Power Losses

Example: Maxim MAX 1947ETA3.3

Setup: We have to supply a sensor module with 3.3 V and 330 μW . The input voltage is $\sim 0.9\text{ V}$ (e.g. from 2 solar cells).

Question: How is the „light load“ efficiency?

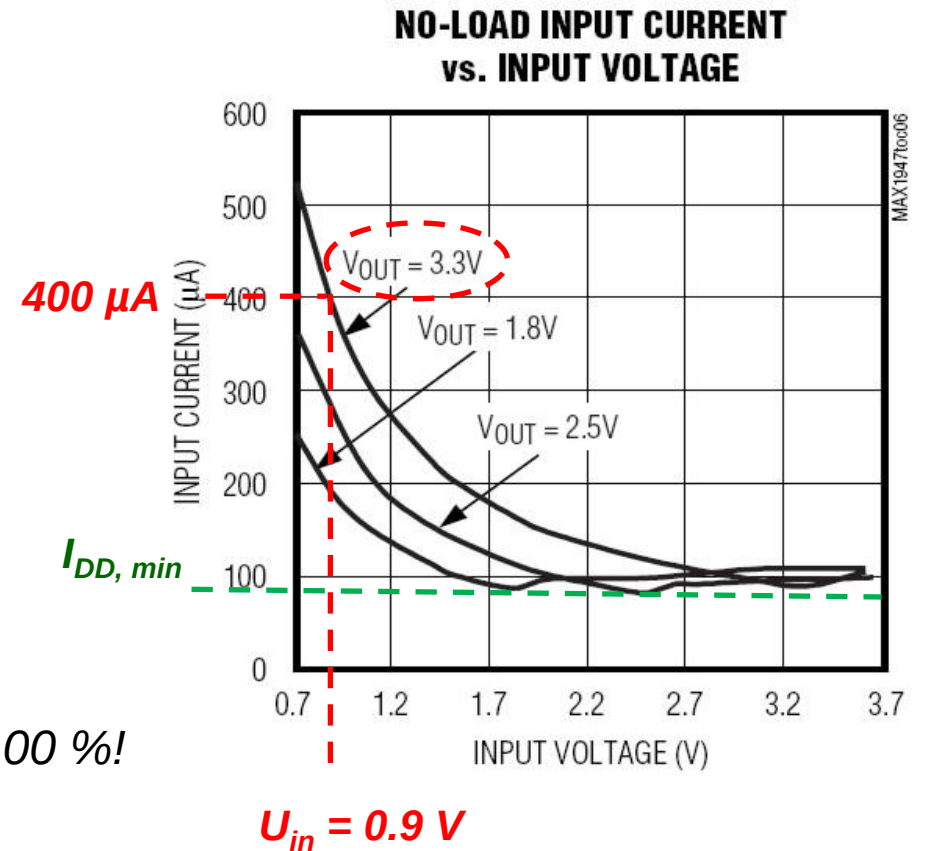
internal losses: $P_S = U_{in} \cdot I_{DD} = 0.9\text{ V} \cdot 400\text{ }\mu\text{A} = 360\text{ }\mu\text{W}$

input power: $P_{in} = P_S + P_{out} + P_{conv} = 690\text{ }\mu\text{W} + P_{conv}$

$$\eta = \frac{P_{out}}{P_{in}} = \frac{330\text{ }\mu\text{W}}{690\text{ }\mu\text{W} + P_{conv}} = 48\% - X\%$$

Efficiency of the step-up conversion itself P_{conv} is unknown but not 100 %!

➔ Result: small efficiency at light load!



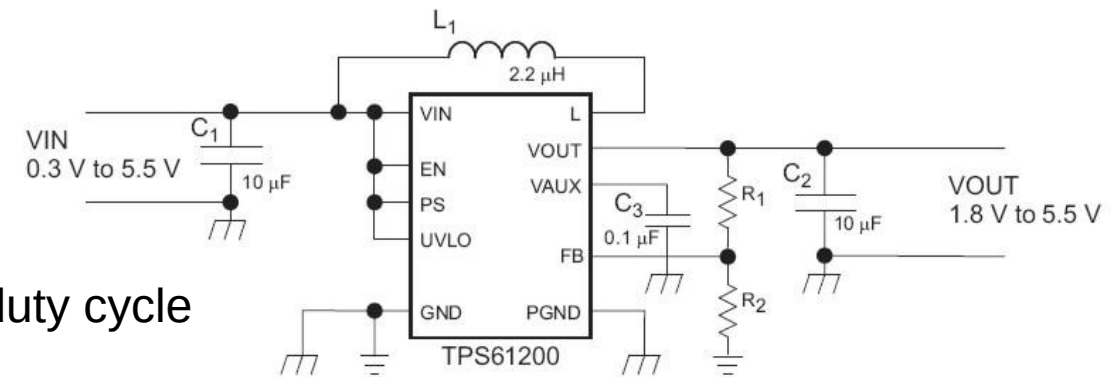
Low-voltage Step-up Converters

Tailored „true“ Low-voltage Step-up Converters

- use of e.g. a low-voltage charge pump to provide the internal supply voltage U_{DD} from low U_{in}
 - ➔ lower internal losses during start-up, lower start-up voltage
- optimized control schemes
 - variable frequencies, peak currents, combined voltage and current regulation, duty cycle modulation, ...*
- *in any case*: synchronous rectification
- low-power CMOS circuitry, MOSFETs with low on-resistance and gate capacitance

Example: Texas Instruments TPS 61200

- start-up voltage: 0.5 V
- minimum operation voltage: 0.3 V (after start-up)
- combined regulation of output voltage, peak current and duty cycle
- system control powered internally from U_{out} and U_{in}



Low-voltage Step-up Converters

Tailored „true“ Low-voltage Step-up Converters

Example: *Texas Instruments TPS 61200*

- shutdown current: $I_{SD} = 1.5 \mu A$
- min. supply current: $I_{DD,min} = 78 \mu A$ (see datasheet)

Setup: We have to supply a sensor module with 3.3 V and 330 μW .

The input voltage is 0.9 V (e.g. from 2 solar cells)

Question: How is the „light load“ efficiency ?

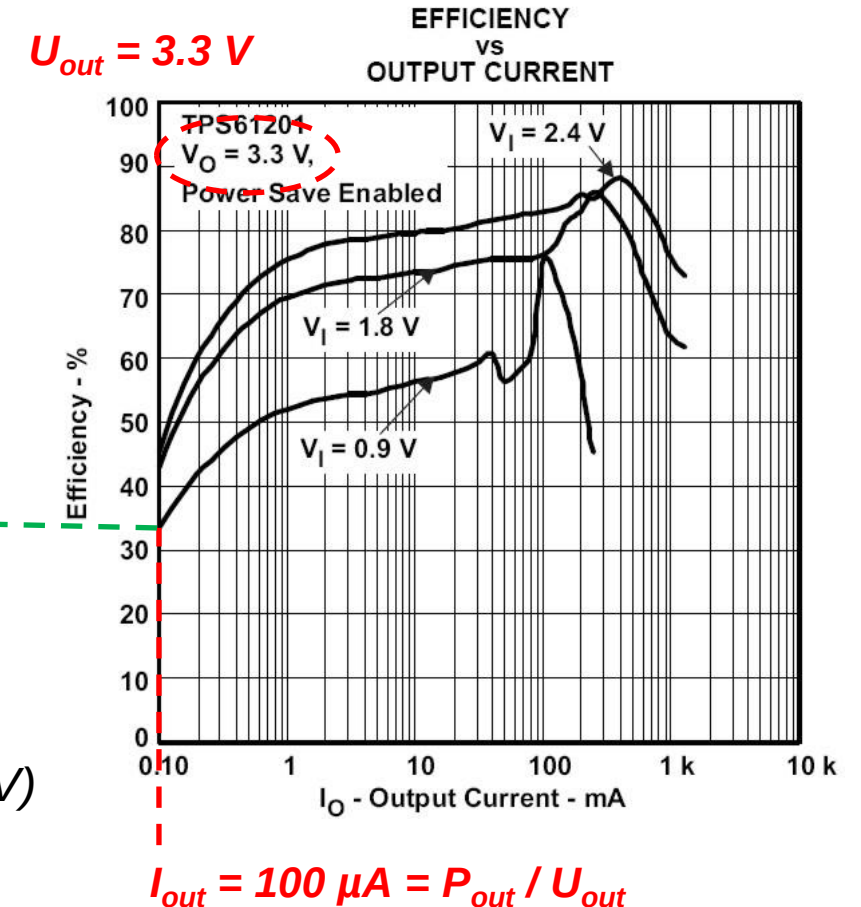
from the datasheet: $\eta = \frac{P_{out}}{P_{in}} = 34 \%$

input power: $P_{in} = \frac{P_{out}}{\eta} = 971 \mu W$

internal losses: $P_S = (1 - \eta) \cdot P_{in} = 640 \mu W$

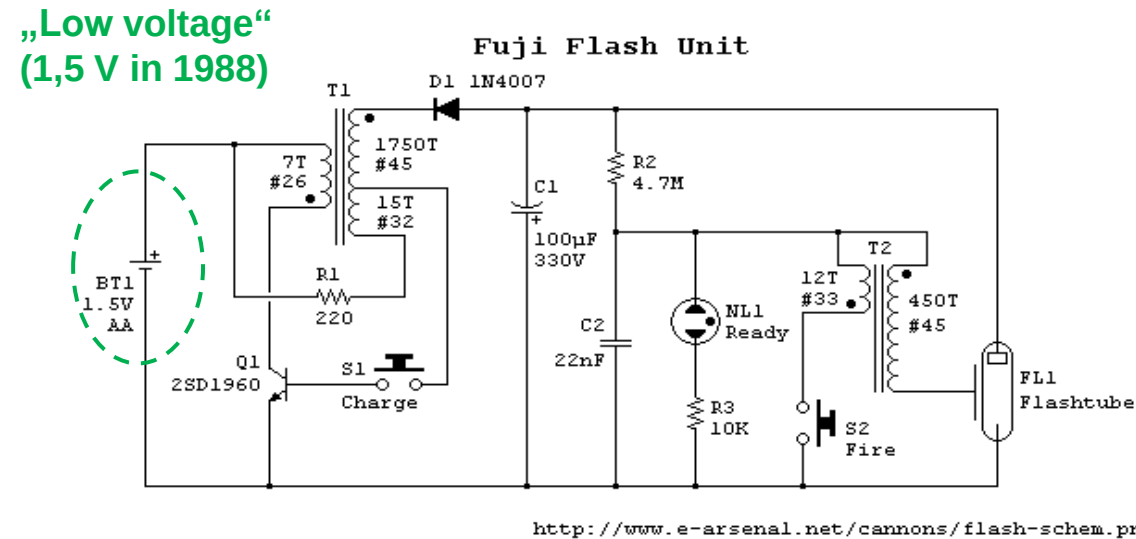
Worse efficiency (?) at light load, BUT much lower start-up voltage (0.5 V)

→ Result: reasonable efficiency at light load!



Resonant Ultra-low-voltage Step-up Converters

Historic Example - 1-cell Photoflash Converter

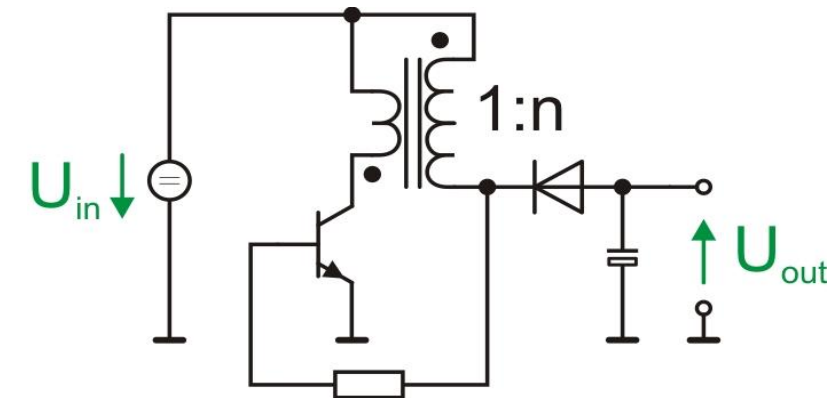


Fuji Quicksnap flash (1988) product flyer

Principle:

„Meißner-Oscillator“

- self-resonant transistor-transformer oscillator (internal feedback with a phase shift of 360°)
- linear or non-linear AC output signal
- rectification of the transformer's output voltage
- up-conversion ratio defined by the transformer turns ratio $n:1$



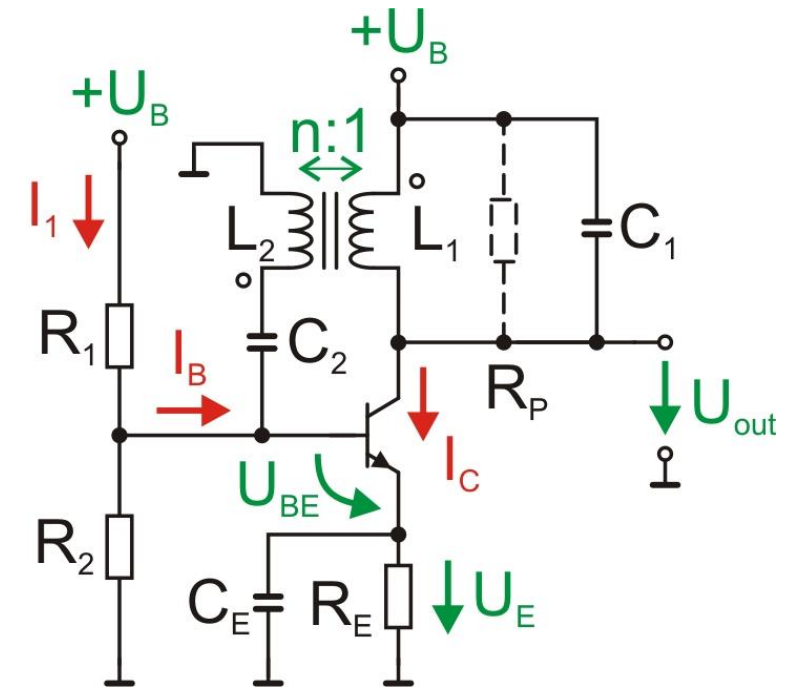
Resonant Ultra-low-voltage Step-up Converters

Linear Meissner Oscillator

- applied for high frequency oscillation
- sinusoidal output signal U_{out}
- precisely adjustable resonance frequency

Design measures and parameters for **linear** operation

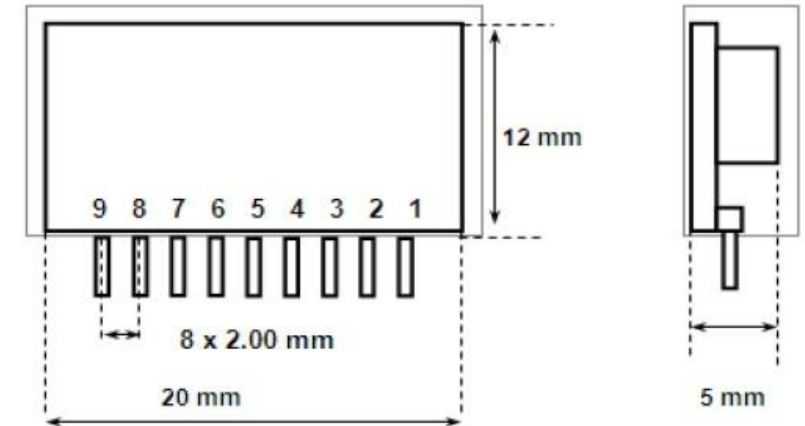
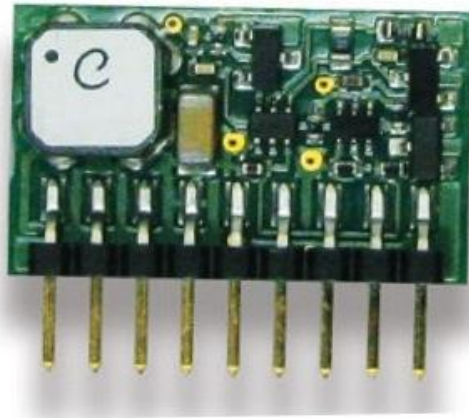
- relatively low turns ratio $1:n$ of the transformer (to obtain linear operation)
 - ➔ no „really high“ output voltage
- definition of the output frequency via resonance frequency of the LC tank
- definition of the output voltage via the quality factor of the oscillator



Resonant Ultra-low-voltage Step-up Converters

Commercial ULV Step-up Module

Enocean
Perpetuum
ECT 300



© Enocean

Physical dimensions: 20x12x5mm
 Typ. input voltage: 0.02 – 0.5V
 Max. input voltage: 1V
 Typ. source impedance: <20ohm
 Typ. output voltage: 3,4V
 Average efficiency: 30%
 Operating temperature: -20°C – 65°C
 Storage temperature: -40°C – 85°C

Pin	Name	Description
1	V _{IN1}	Voltage input 1 connection to Peltier/ Seebeck element
2	NC	Not connected
3	NC	Not connected
4	V _{IN2}	Voltage input 2 connection to Peltier/ Seebeck element
5	V _{OUTAC}	Voltage output AC (internal test only)
6	NC	Not connected
7	V _{OUT}	Voltage output for load
8	NC	Not connected
9	GND	Ground

Resonant Ultra-low-voltage Step-up Converters

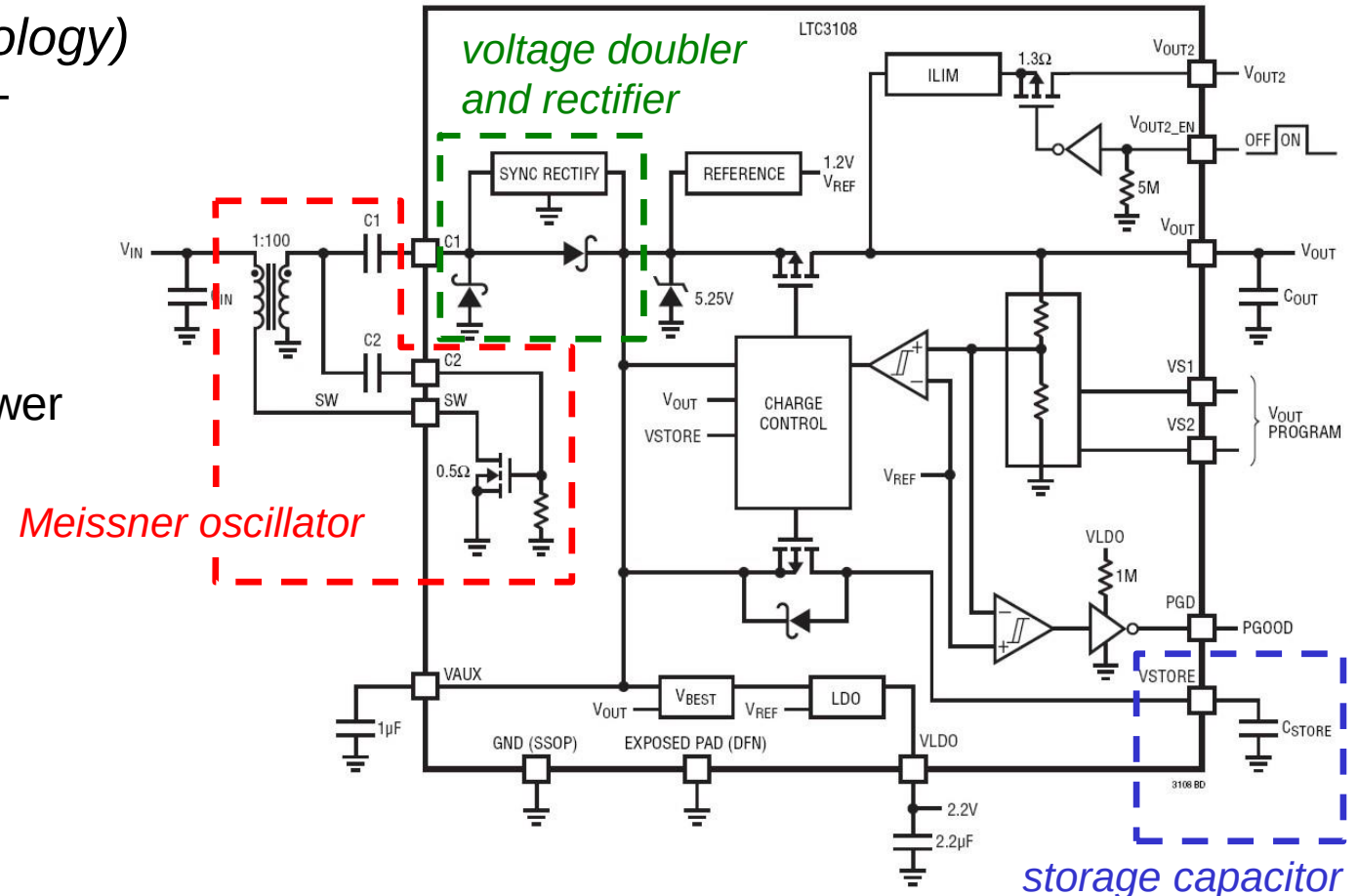
Commercial Monolithically Integrated Version

Example: LTC 3108/3109 (Linear Technology)

- Meissner oscillator with depletion MOSFET (instead of JFET)
- capacitive voltage doubler with synchronous rectification
- internal power management system for power distribution to/from a storage capacitor

Technical data

- start-up voltage: 20 mV
- typical efficiency: 35% at 20 mV



Power Conversion

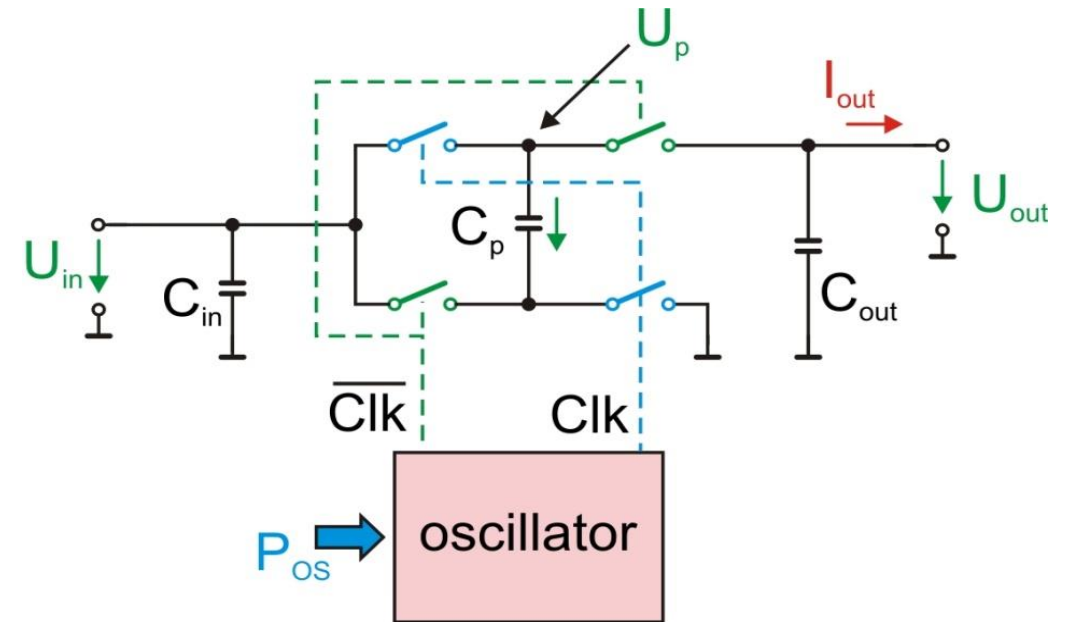
Charge Pump Converters (Capacitive Converters)

Function principle: Two pairs of switches are operated with opposite clocks to charge and discharge a “pump capacitor” C_p .

- Cycle 1: C_p is charged to V_{in}
- Cycle 2: C_p serially connected to V_{in} , both parallelly connected to C_{out} .

Properties

- main losses are switching losses
- voltage doubling and voltage inversion possible
- advantageous for small power systems
- optimum efficiency for pure „voltage doubling“, reduced efficiency for other voltage ratios



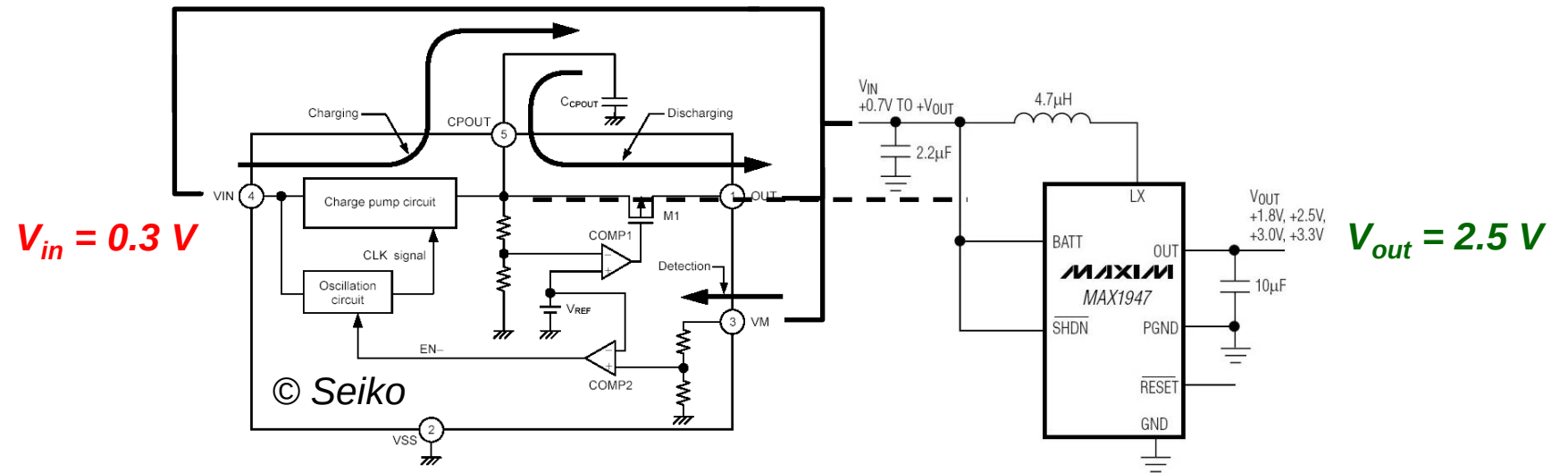
$$U_{out} = \frac{C_p}{C_p + C_{out}} \cdot 2 \cdot U_{in}$$

Power Conversion

Charge Pump Converters (Capacitive Converters)

Example: Seiko S 883 (here in combination with a step-up converter MAX 1947)

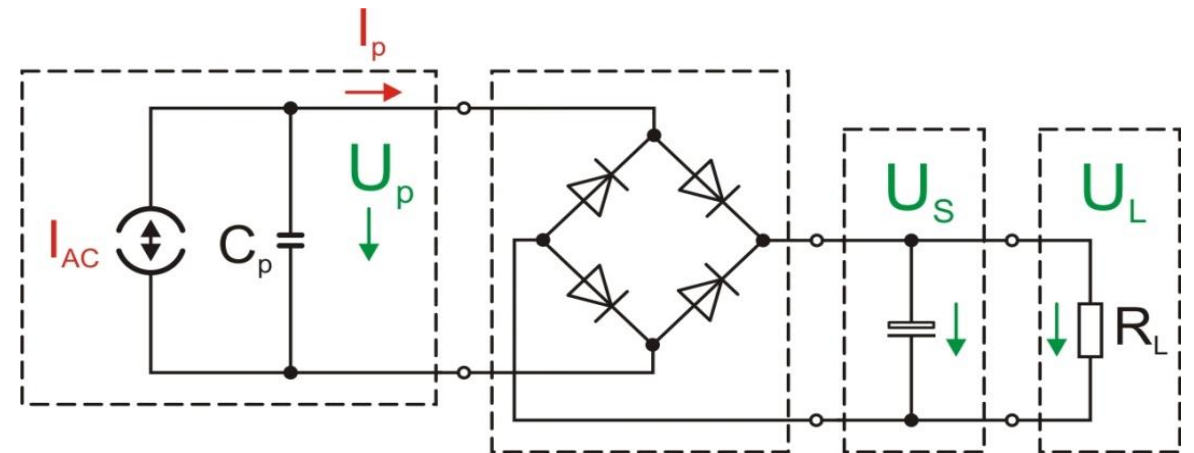
- uses SOI (silicon on insulator) transistors with low-threshold voltages ➡ low start-up voltage (300 mV)
- intended as a start-up circuit for an inductive step-up converter
- output capacitor C_{out} of charge pump is filled and used as short-term power source for a “kick start”
- boost converter starts oscillating and is supplied by its own output
- charge pump is turned on (over VM) and turned on again, if required.



Energy Extraction from Piezoelectric Generators

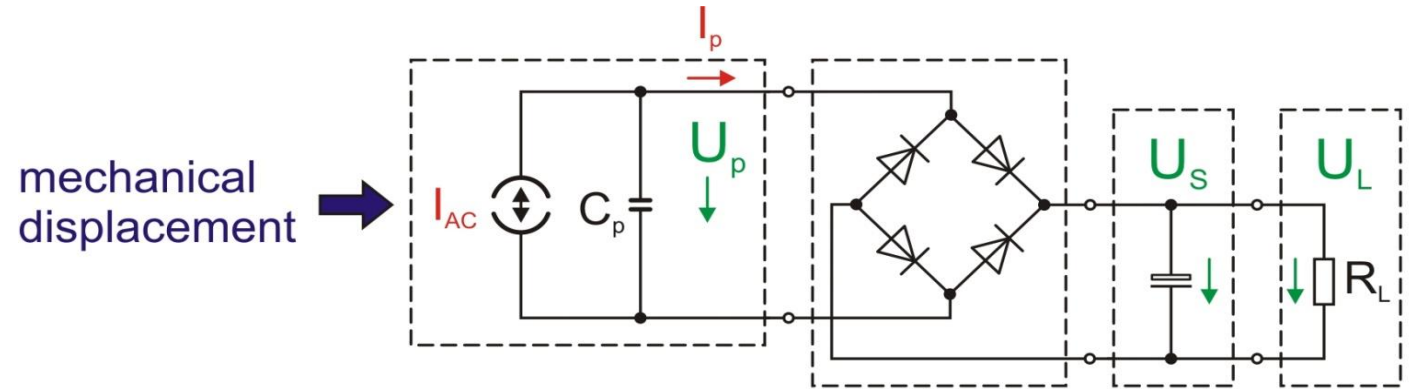
General

- Piezogenerators provide AC currents
- rectification required
- state of the art: bridge rectifiers with low efficiency
- capacitive nature of the generator
 - ➔ phase shift between current and voltage
 - ➔ maximum of voltage $\neq$ maximum of current
 - ➔ When to extract power?



Energy Extraction from Piezoelectric Generators Via Bridge Rectifier

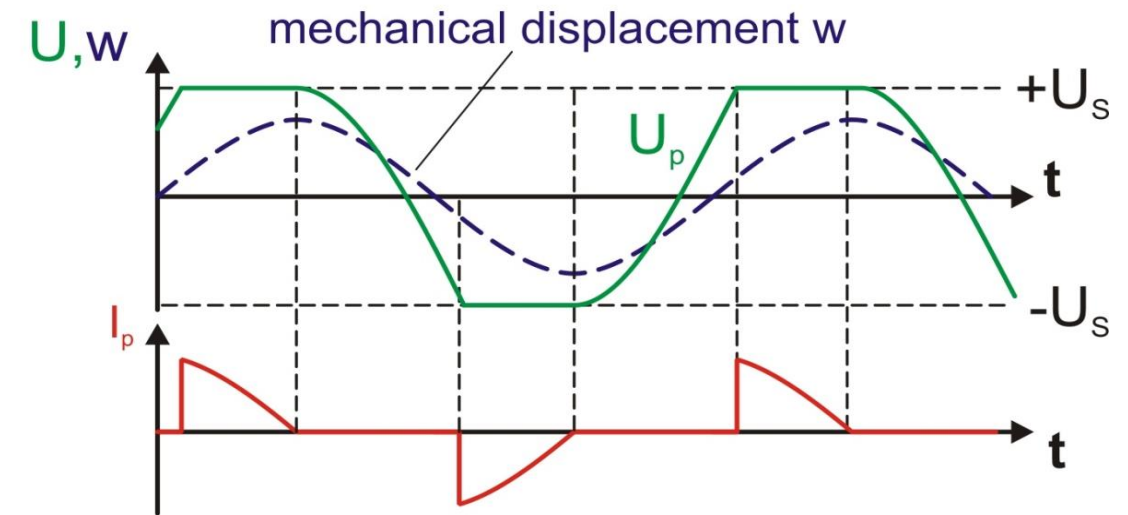
Voltage and current situation



Consequences

- piezogenerator can only deliver charge, when $|U_p| > U_s$
- phase shift between voltage and current (depending on the values of C_p and R_L)

➡ no ideal energy transfer



E. Lefeuvre et al., Sensors and Actuators A 126, 2006, 405–416

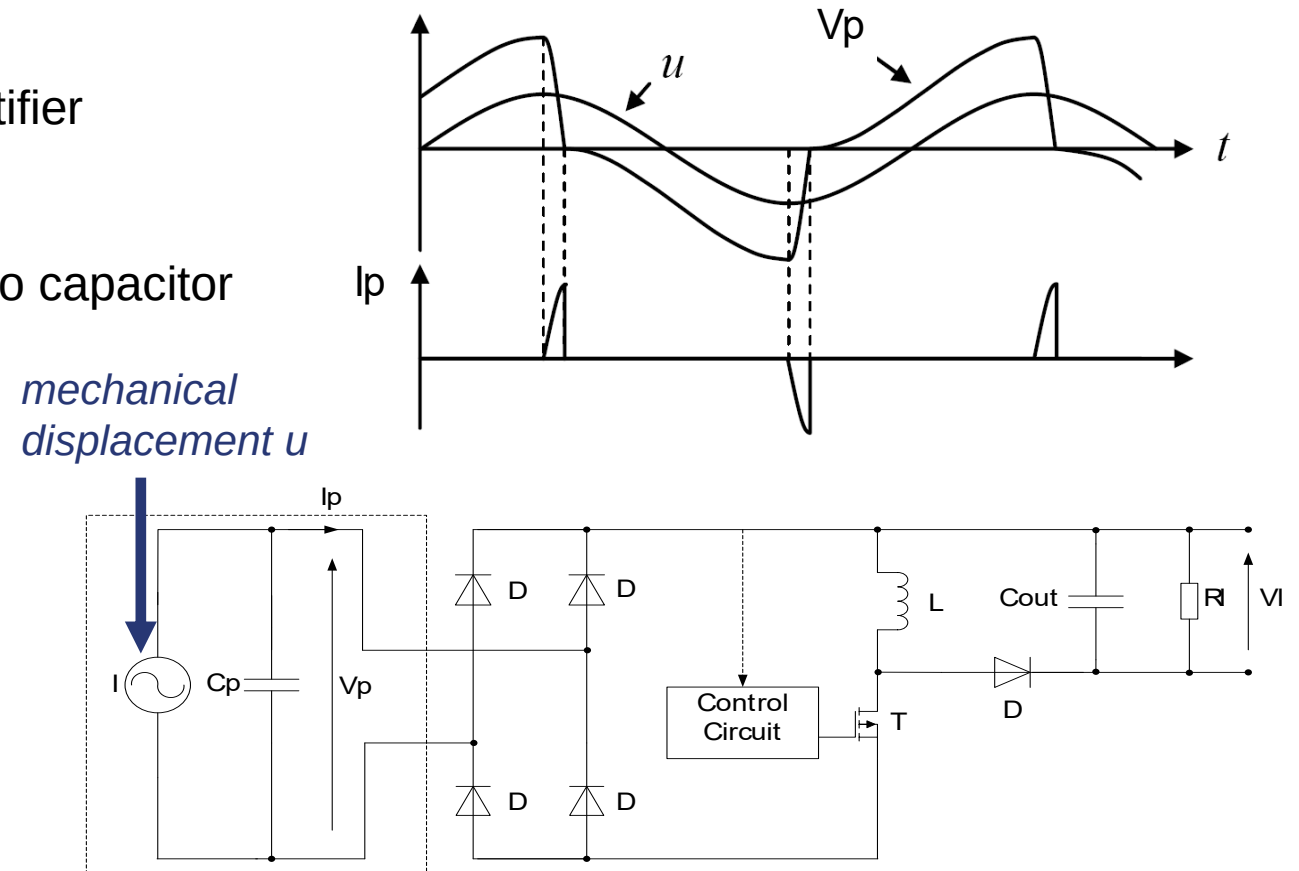
Energy Extraction from Piezoelectric Generators

Synchronous Electric Charge Extraction (SECE)

- combination of step-up converter and bridge rectifier
- control circuit detects peak value of V_p
- transistor T is turned on and discharges the piezo capacitor
- step-up conversion of the current pulse

Consequences

- current and voltage are „shifted“ into phase
 - extraction at the power maximum
 - challenge: peak detection
- ➡ nevertheless: improved energy transfer



E. Lefeuvre et al., Sensors and Actuators A 126, 2006, 405–416

Energy Extraction from Piezoelectric Generators

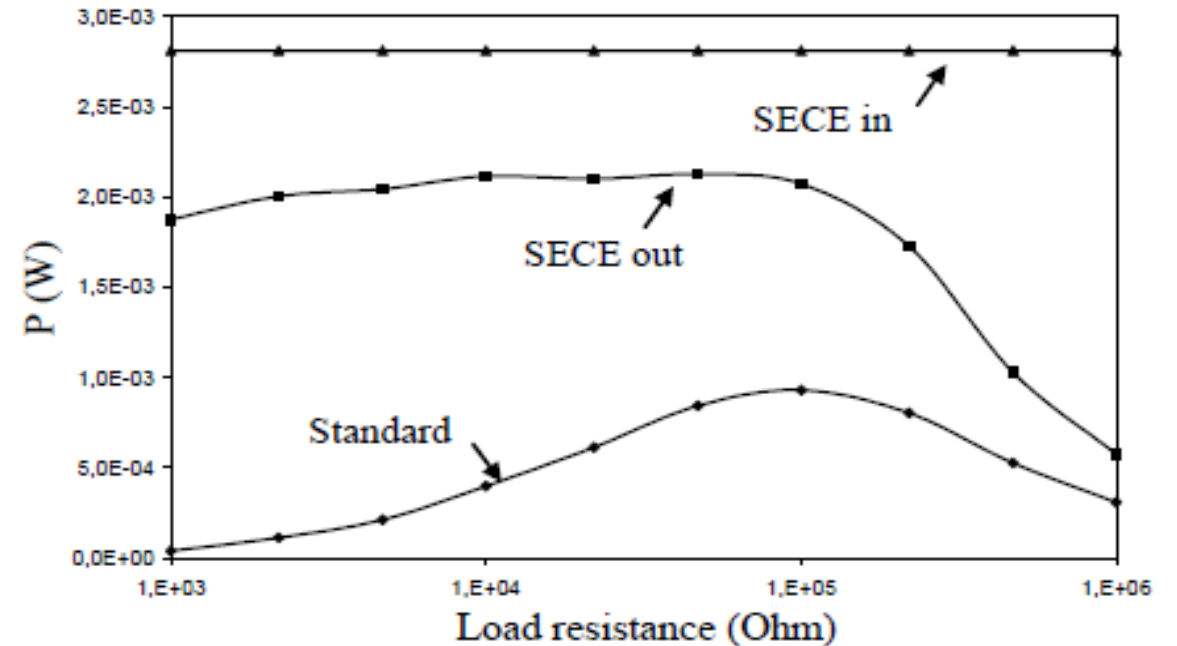
Synchronous Electric Charge Extraction (SECE)

Comparison of conversion efficiencies

Results

- SECE in: flat power curve
- Bridge (Standard): output power depends on load resistance
- SECE out: More than 100% improvement in power output over bridge rectifier

➡ extraction efficiency up to 80%



See also: Gasnier et al, IEEE JOURNAL OF SOLID-STATE CIRCUITS, VOL. 49, NO. 7, JULY 2014

E. Lefeuvre et al., Sensors and Actuators A 126, 2006, 405–416

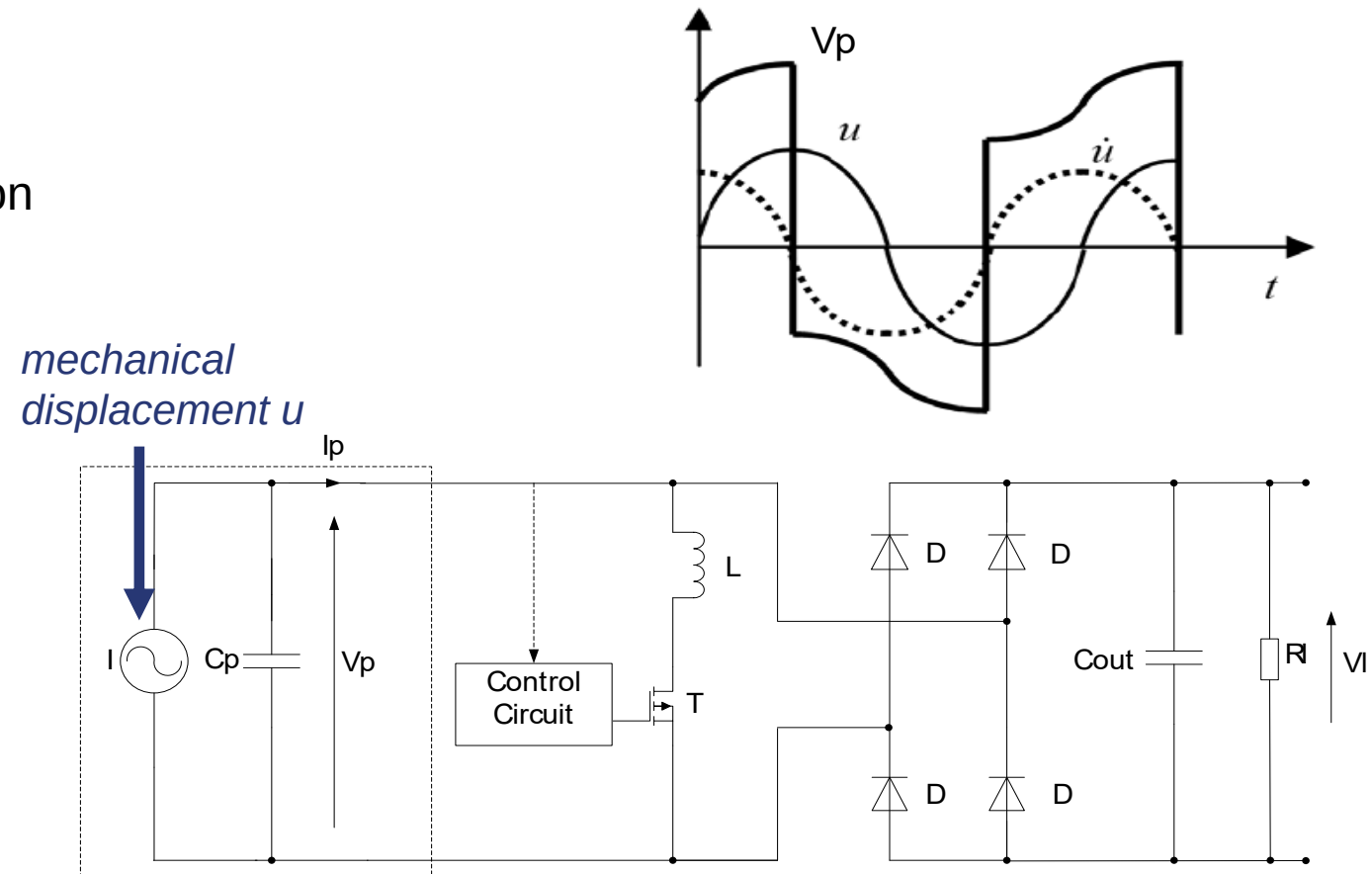
Energy Extraction from Piezoelectric Generators

Synchronized Switch Harvesting on Inductor (SSHI)

- similar to SECE, however...
- step-up conversion happens **before** rectification
- higher efficiency

Consequences

- current and voltage are „shifted“ into phase
 - extraction at the power maximum
 - challenge: peak detection
- ➡ **nevertheless: improved energy transfer**



A. Badel et al., J. Intell. Mat. Syst. Struct. 16, 2005, 865-876.

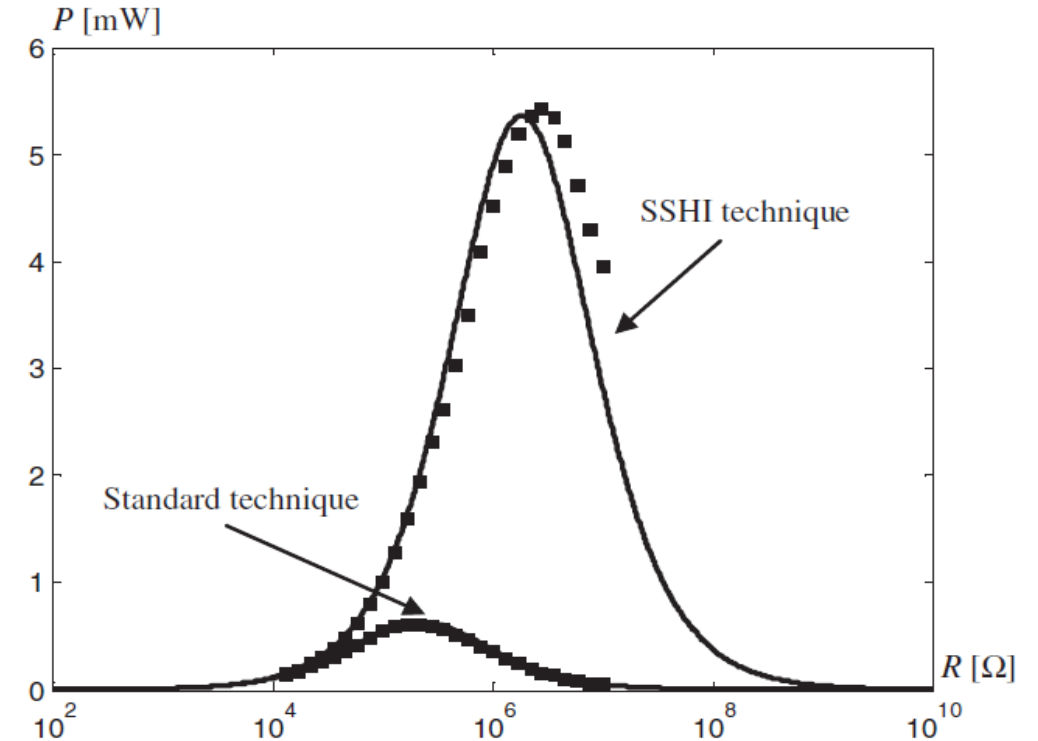
Energy Extraction from Piezoelectric Generators

Synchronized Switch Harvesting on Inductor (SSHI)

Comparison of conversion efficiencies

Results

- Both concepts (SSHI and bridge) perform an optimum extraction for specific load resistances
- SSHI: more than 600 % improvement in power output over bridge rectifier



A. Badel et al., J. Intell. Mat. Syst. Struct. 16, 2005, 865-876.

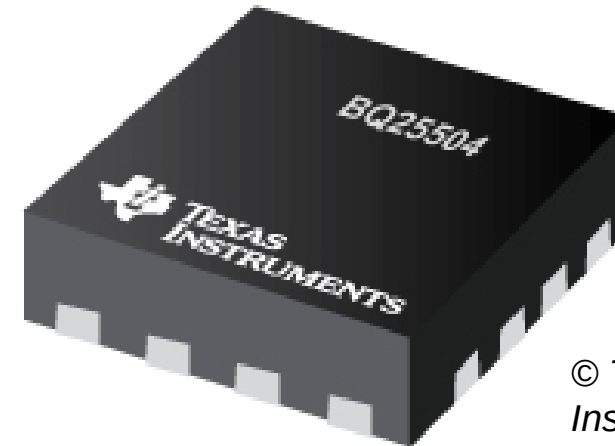
Energy Harvesting Power Management ICs

Examples

Texas Instruments TI BQ25504

Ultra low power DC/DC boost converter with...

- nano-power management: *ultra-low quiescent (self supply) current: $I_Q < 330 \text{ nA}$*
 - programmable MPPT *for optimal energy extraction, works for different generator types*
 - low-input while working $V_{IN} \geq 80 \text{ mV}$
 - low cold-start voltage $V_{IN} \geq 330 \text{ mV}$
 - adjustable over/under-voltage
- for different types of storages:
- Thin-film
 - Caps
 - Super-caps
 - Li-ion batteries
- BQ25505 with integrated setup only for Li-Ion batteries



© Texas
Instruments

Energy Harvesting Power Management ICs

Examples

Linear Technology LTC 3588

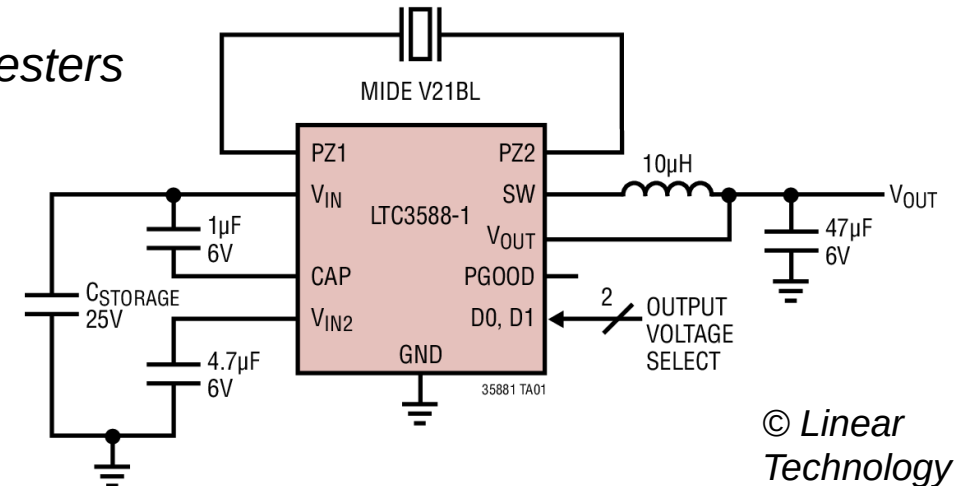
Nanopower energy harvesting power supply for piezoelectric harvesters

- wide input voltage range $V_{IN} = 2.7 \dots 20V$
- integrated low-loss full-wave bridge rectifier
- high efficiency integrated hysteretic buck DC/DC
- selectable output voltages of 1.8V, 2.5V, 3.3V, 3.6V

Other devices from LT

- LTC 3106 - *Low Voltage Buck-Boost Converter* (solar)
- LTC 3107 - *Ultra-Low Voltage Energy Harvester and Primary Battery Life Extender* (thermal and solar)
- LTC 3108 - *Ultra-Low Voltage Step-Up Converter and Power Manager* (thermal)
- LTC 3109 - *Auto-Polarity, Ultra-Low Voltage Step-Up Converter and Power Manager* (thermal)

100mA Piezoelectric Energy Harvesting Power Supply



Energy Harvesting Power Management ICs

Examples

ST Microelectronics SPV 1050

ultra-low-power power management IC

- for thermoelectric generators and PV modules
- low self-supply current: $I_Q = 1 \dots 2.8 \mu A$
- low input voltage, when working: $V_{IN} \geq 150 \text{ mV}$
- low cold-start voltage: $V_{IN} \geq 500 \text{ mV}$
- resistor-programmable MPPT: *optimal energy extraction from different generators*
- integrated battery management with selectable charge and discharge voltage levels
- integrated storage capacitor management
- integrated power distribution with battery disconnect
- two linear low-drop voltage regulators



© ST Microelectronics

Conclusion

- Power shaping and voltage conversion is necessary to deal with the different output properties of the variety of harvester
- Most state-of-the art power management circuits are not well suited for energy harvesting.
- Low-voltage circuits are required for several generators and - almost - unavailable.
- Different methods for more efficient energy extraction from harvesters are available, yet in research state
- Several “old” principles from the “single-transistor” age perform well for energy harvesting, see the more recent ULV step-up converters.
- Matched power extraction is required and offers significant gains in power.
- A lot of voltage conversion IC’s are available from different companies
- Some are already pretty application specific, which might be useable directly
- Not all of them work as good as the datasheet makes you believe...